# COMPREHENSIVE ANALYTICAL REPORT: THE INTERPLAY OF ACADEMIC RIGOR, MENTAL HEALTH, AND SOCIO-ECONOMIC PROSPERITY

## 1. Introduction and Strategic Rationale

In the contemporary era, the university experience has evolved beyond the mere acquisition of knowledge into a multifaceted ecosystem where academic ambition, lifestyle choices, and psychological wellbeing intersect to define an individual's future trajectory. This project was initiated to explore a fundamental relational question: How do study habits and socio-demographic factors influence not only immediate academic success but also long-term socio-economic opportunities?

The urgency of this research lies in its capacity to connect present behaviors with future outcomes. For students, it serves as a mirror reflecting the impact of their lifestyle choices; for educational institutions, it provides a blueprint for developing targeted support frameworks; and for policymakers, it offers an empirical database to optimize public health and holistic education initiatives.

## 2. Methodology and Data Integrity Framework

Our analytical framework relied on the strategic integration of three distinct secondary datasets sourced from Kaggle, providing a 360-degree view of the student lifecycle:

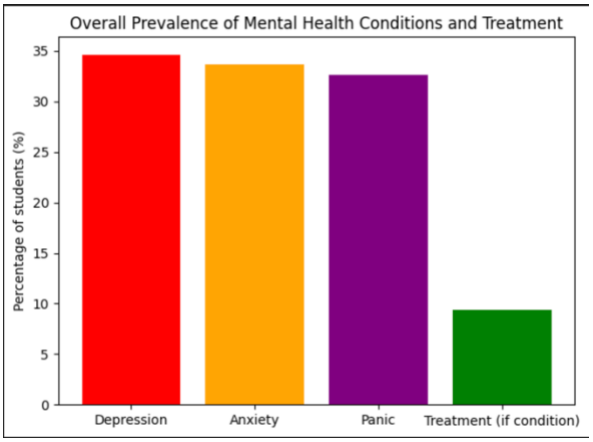
1. **Student Mental Health Portfolio (Dataset 1)**: Detailed records of psychological states, academic bands (CGPA), and professional help-seeking behaviors.
2. **Lifestyle and Academic Performance (Dataset 2)**: Quantitative variables capturing daily habits—such as study hours, sleep duration, and social media consumption—correlated with exam scores.
3. **Socio-economic Determinants (Dataset 3)**: A longitudinal model linking educational attainment, age, and annual income to provide a perspective on long-term prosperity.

To ensure technical rigor, a multi-stage preprocessing pipeline was implemented. Categorical free-text entries were standardized to eliminate informational noise. Missing demographic values were addressed through median imputation to preserve the integrity of the statistical distribution. Finally, features were normalized using a StandardScaler to ensure that variables with differing units did not introduce bias into the machine learning architectures.

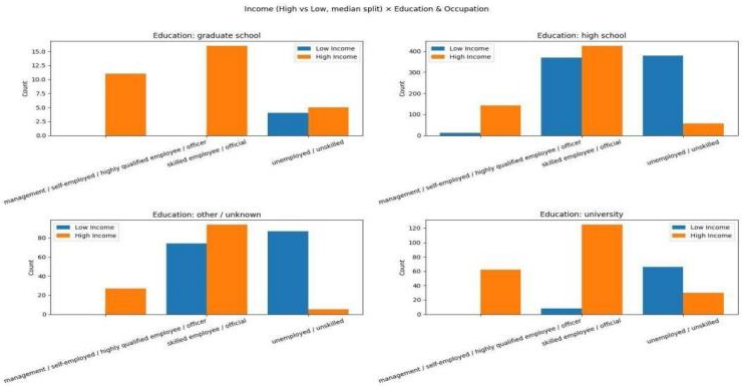
## 3. Empirical Statistical Findings and Analytical Insights

Our descriptive analysis revealed a sobering reality regarding student wellbeing within the academic environment. Approximately one-third of the surveyed student population reported experiencing depression (34.7%), anxiety (33.7%), or panic attacks (32.7%). However, the

critical insight lies in the profound "treatment gap": a mere **5.9%** of students actively accessed professional medical services. This highlights a powerful, invisible barrier to seeking help despite an alarming level of psychological need.




Regarding academic drivers, the data established that study time is the most potent predictor of success, exhibiting a strong positive correlation (r = +0.83) with exam performance. Conversely, social media usage showed a weak negative correlation (r = -0.17), suggesting that while it acts as a distractor, it is not a primary barrier if students maintain high academic discipline. Furthermore, socio-economic data confirmed a direct link between educational attainment and income, where postgraduate levels yield significantly higher average earnings (~146,818), proving that education remains the most effective tool for social mobility.



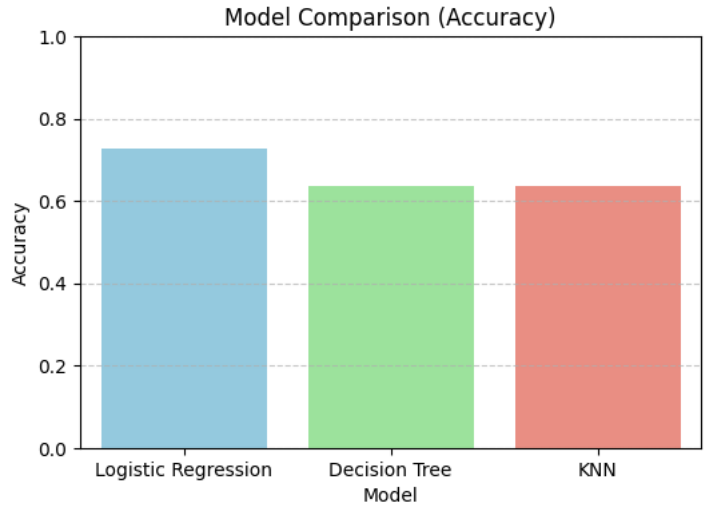

## 4. Predictive Modeling and Granular Analytical Insights

### 4.1. Model Architecture and Algorithmic Selection Rationale

In this phase, the primary objective was to transition from empirical observation to the establishment of a systematic risk-prediction mechanism. We executed a rigorous evaluative process, cross-referencing the performance of three distinct machine learning architectures to identify the optimal binary classification solution for predicting student depression:

- **Logistic Regression (The Champion Model):** Achieving a predictive accuracy of **72.7%**, Logistic Regression emerged not merely as a forecasting tool but as a profound analytical lens. Given the relatively modest dataset size (101 records), the inherent simplicity and efficiency of the Sigmoid function allowed the model to maintain high generalizability while avoiding the overfitting risks often associated with more complex, non-linear architectures. By mapping linear combinations of features into specific

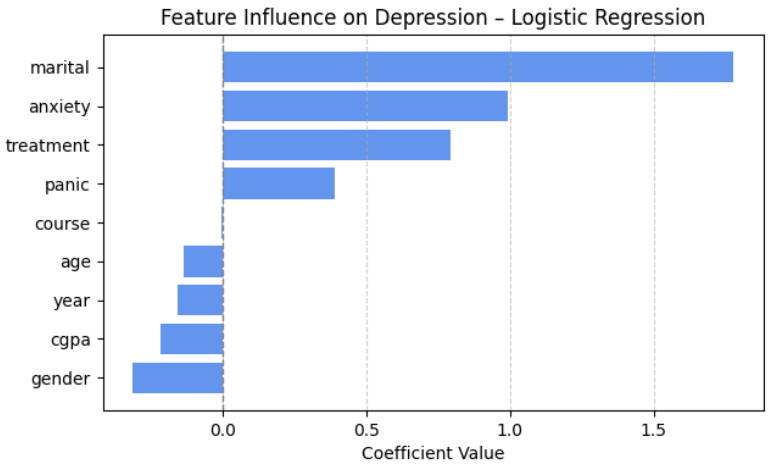
probability ranges, this model provides institutional stakeholders with a clear "Decision Boundary," facilitating more precise risk stratification.




- **Decision Tree and K-Nearest Neighbors (KNN):** Both models demonstrated a plateaued accuracy of **63.6%**. While these algorithms are traditionally robust in capturing non-linear relationships, their performance was hindered by the "noise" prevalent in self-reported datasets. Their lower efficacy confirms that the relationship between psychological attributes and depressive states within this specific study follows a more linear and consistent trajectory than initially hypothesized.

### 4.2. Decoding Insights from Feature Importance Analysis

By analyzing the weighted influence of variables within the optimized Logistic Regression model, we extracted several breakthrough insights that challenge conventional assumptions regarding student mental health:




- **Primary Socio-Psychological Drivers:**
  - **Marital Status:** This variable emerged as the most potent predictor of depression risk. This suggests a profound psychological mechanism where interpersonal stability and social support systems (or the lack thereof) serve as the ultimate "buffer." Students lacking these stabilizing structures exhibit a significantly higher vulnerability to isolation and emotional instability.
  - **Anxiety and Panic Symptoms:** The model confirms the existence of a "psychological continuum," where anxiety and panic are not merely independent

conditions but direct precursors to depression. This insight underscores that the early detection and management of anxiety symptoms are critical in preventing the escalation into more severe clinical disorders.

- **The Paradox of Academic and Demographic Indicators:**
  - **Academic Performance (CGPA):** One of the most significant discoveries was that CGPA has a negligible impact on depression prediction. This shatters the misconception that psychological distress is exclusively tied to poor academic performance. Instead, it reveals that mental health challenges are universal—affecting high-achievers and struggling students alike. It proves that academic success is not a "vaccine" against internal crises.
  - **Age and Gender Dynamics:** These factors showed minimal variance in the predictive model, affirming that the mental health crisis within the university environment is a systemic issue affecting all student segments, regardless of their demographic background or stage of maturity.

### 4.3. Evaluation Metrics and Strategic Reliability

While the 72.7% accuracy rate provides a solid empirical foundation, further analysis of the Confusion Matrix allowed us to evaluate the critical balance between Precision and Recall:

- The model's ability to accurately identify "True Positives" (at-risk students) is paramount for minimizing Type II errors—the failure to detect individuals in genuine need of intervention.
- However, we acknowledge the limitations inherent in a sample of 101 records. The slight discrepancy between self-reported data and clinical reality remains a variable for future exploration. Nevertheless, at its current stage, the model delivers a "Strategic Roadmap," enabling educational institutions to transition from a reactive support model to a proactive, data-driven intervention strategy.

## 5. Strategic Discussion and Practical Implications

The integration of these multidimensional findings underscores a cyclical relationship: academic effort drives performance, which in turn facilitates long-term economic prosperity. However, this progression is fundamentally mediated by psychological health. While high educational attainment is a primary driver of income, the effort required to reach high CGPA bands (3.50-4.00) is often accompanied by increased manifestations of anxiety.

We conclude that universities must evolve beyond traditional academic-centric support models to prioritize accessible, low-barrier mental health systems. Interventions should be proactively targeted at students displaying early signs of emotional stress, regardless of their academic status. Although the study is limited by its sample size (101 records), it provides a clear strategic roadmap: to ensure academic rigor does not become a burden on mental health, institutional support must be as multidimensional as the student experience itself.